MACRO Environmental Issues

Ozone Layer Depletion

- Pioneering investigations by Sherwood Rowland and Mario Molina in early seventies (1974).
- Soon it was established to be ‘true’
- Our atmosphere consists of :
  - Troposphere (0 to 15km)
  - Stratosphere (extends up to 60 km)
  - Mesosphere
  - Thermosphere

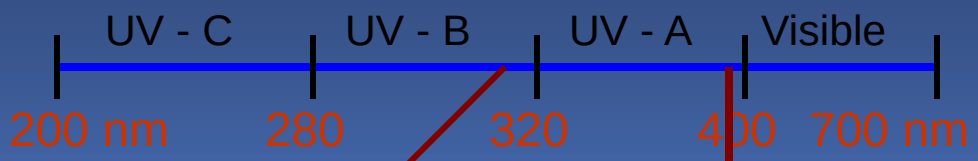
- O_3 in troposphere is a pollutant,
  - helps in formation of smog,
  - capable of rapidly removing vitamin E from skin
  - Can cause corrosion problems
  - Damage vegetation
  - Can cause mutation in DNA

When troposphere ozone layer reaches 50ppb, plant growth reduces by about 15%

Ozone in the stratosphere

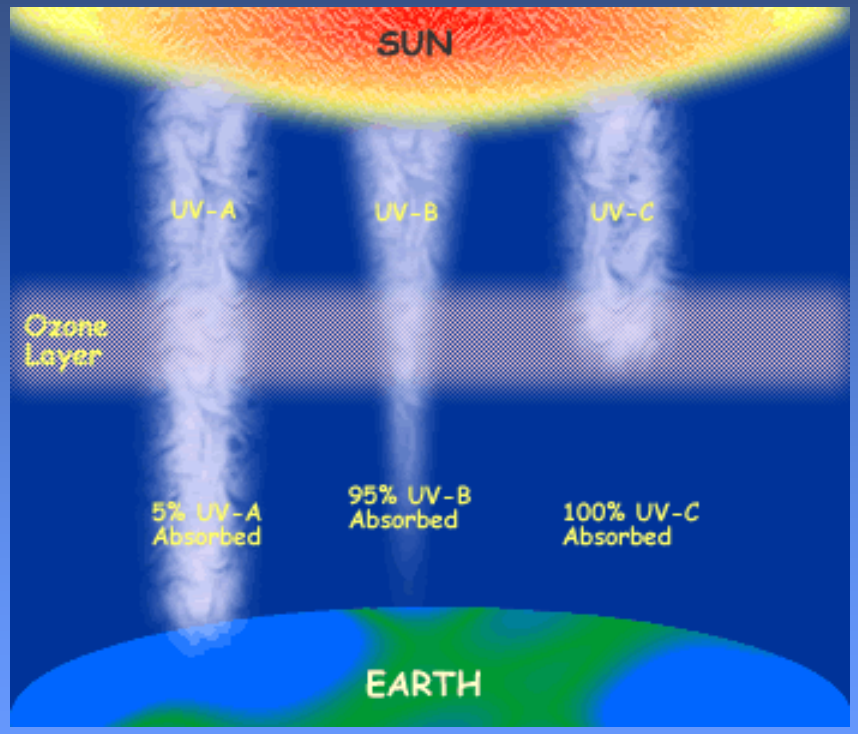
- O_3 in the stratosphere is a blessing as it shields the passage of dangerous solar UV-B radiations to the earth.
- O_3 absorbs these radiations (range 280 – 310 nm)
- UV-B radiations can cause skin cancer

Effect of UV radiation in Troposphere



Reddening of skin – Erythema
Reduction in Vitamin – D synthesis
Skin cancer,
Cataracts, and
Damage to the immune system
Slowed plant growth

Synthesis of Vitamin – D



Ozone Layer depletion

- Patches in the Stratosphere where the Concentration of Ozone is very Low
- In fact O_3 is continuously formed in natural processes
- But the rate of formation is considerably lesser than the rate of destruction
- CFCs are the dominant source of Chlorine which destroys the ozone molecules

Natural reactions

- O_3 + UV light → O + O_2
- O + O_2 → O_3

Reactions when CFC is there

- Cl_2CF_2 + UV light → $ClCF_2$ + Cl
- Cl + O_3 ---→ ClO + O_2
- ClO + O_3 --→ Cl + O_2
- Cl is regenerates and it keeps on reacting with O_3

Ozone Layer Depletion

- Chlorine is regenerated in the reaction. Theoretically one chlorine atom can destroy infinite Ozone molecules (about 1,00,000)
- As a result, global ozone amount was decreasing @ 1% per year (reports in mid nineties)
- 1% per depletion leads to 1.3 to 1.8% increase of UV radiations reaching earth's surface.

Amount of O_3 in the atmosphere

- It varies geographically and seasonally
- It is less than about 260DU in equatorial latitudes and about 400 DU at the poles
- It is less in Sept and more in Dec (with a difference of about 150 DU)

DU = Dobson unit

1 DU = 2.69×10^{20} O_3 molecules / m^2 air column

300 DU corresponds to 3 cm thick layer

Distribution of Ozone in Stratosphere

In the 1920s, G.M.B. Dobson developed a spectrometer that could measure small concentrations of ozone.

If vertical column of Ozone is brought to surface atmosphere at 0 ^{O}C, its thickness will be 4cm = 400 DU

1 cm = 100 Dobson Unit
(After Dr. D.M.B. Dobson)

The Culprits (ODS – Ozone Depleting Substances)

- O_3 is depleted by the reactions involving a variety of compounds reaching the stratosphere

Use of halons is banned in many countaries nder Montreal Protocol

- Important ODSs are : CFCs, Halons, CCl_4, etc
- CFC-11 and 12 (Freon 11 and 12) are the most dangerous ones.

CFCs

- They take about 6 years to reach the stratosphere (poles)
- CFCs break down in the stratosphere by UV-radiations releasing Cl.
- The catalytic chain reactions initiated by Cl atoms, involving ClO, deplete the O_3 layer
- ClO is the main destroyer of ozone
- ClO concentration is about 0.5 to 1.0 ppbV

- Polar regions are more prone to O_3 depletion as polar regions collect more dust and CFCs
- In darkness, tiny ice crystals are formed in the polar stratosphere facilitating the O_3 depletion when sun appears to trigger the reactions.

CFCs

- Developed in 1930s as a safe alternative to NH_3 and SO_2 refrigerants
- They were considered to be one of the most useful class of compounds ever developed because of: chemical stability, non-inflammability, non-corrosiveness, low toxicity, etc.
  - In fact inertness & stability creates the problem of ozone layer depletion

- CFCs are accumulating in the stratosphere about 5 times faster than they could be destroyed by UV radiations
- CFC concentration in the atmosphere increased from 0 (pre-industrial era) to 0.88 ppbV (1990 report) at an annual growth rate of 4%.
- Prediction: by 2010: 205 million kg of CFC 11 and 420 million kg of CFC 12.

How Ozone is the destroyed in the Stratosphere

1. The polar winter leads to the formation of the polar vortex which isolates the air within it.

2. Cold temperatures form inside the vortex; cold enough for the formation of Polar Stratospheric Clouds (PSCs). As the vortex air is isolated, the cold temperatures and the PSCs persist.

3. Once the PSCs form, heterogeneous reactions take place and convert the inactive chlorine and bromine reservoir species to more active forms of chlorine and bromine.

4. As sunlight returns to the air inside the polar vortex, the production of active chlorine and catalytic ozone destruction cycle begins. Ozone loss is rapid.

CFC – Lovelock's experiment

From London to North Atlantic, Lovelock found CFC-11 in every air sample he collected. This unexpected discovery prompted Lovelock to do further studies. Accordingly, he asked the British government for a modest sum of money to place his apparatus on board a ship travelling from England to Antarctica. His request was rejected; one reviewer commented that even if such a measurement succeeded, he could not imagine a more useless bit of knowledge than finding the atmospheric concentration of CFC-11.

But Lovelock persisted. Using his own money, he put his experiment aboard the research vessel Shackleton in 1971. Two years later the British researcher reported that his shipboard apparatus had detected CFC-11 in every one of the more than 50 air samples collected in the North and South Atlantic. Lovelock correctly concluded that the gas was carried by large-scale wind motions. He also stated that CFCs were not hazardous to the environment, a conclusion soon to be proven wrong.

CFC – Passage to Stratosphere

In 1972, the life of atmospheric scientist F. Sherwood Rowland took a critical turn when he heard a lecture describing Lovelock's work. Like other researchers at the time, Rowland had no inkling that CFCs could harm the environment, but the injection into the atmosphere of large quantities of previously unknown compounds piqued his interest. What would be the ultimate fate of these compounds? Rowland, joined by Mario Molina, a colleague at the University of California, Irvine, decided to find out.

Most molecules that are propelled into the atmosphere are removed within a few hours to a few weeks by **three** general processes.

1. Photolysis, which is when the molecule absorbs sunlight.
2. Dissolution in water, or rainout.
3. Molecules are removed through a reaction with the hydroxyl radical (HO) or ozone, a process called oxidation.

CFC molecules differ from most molecules in that they are transparent to sunlight, are insoluble in water, and are chemically inert to oxidizing agents. Therefore the aforementioned processes will not remove them from the atmosphere. The molecules can remain in the lower atmosphere for a long time before being pushed into the stratosphere by powerful storms at the equator

ODS – Research of Molina and Rowland

In examining these fragments, Rowland and Molina were aided by prior basic research on chemical kinetics--the study of how quickly molecules react with one another and how such reactions take place. Scientists had demonstrated that a simple laboratory experiment will show how rapidly a particular reaction takes place, even if the reaction involves the interaction of a chlorine atom with methane at an altitude of 18 miles and a temperature of -60 ºF.

Rowland and Molina did not have to carry out even a single laboratory experiment on the reaction rates of chlorine atoms. They had only to look up the rates already measured by other scientists. Basic research into chemical kinetics had reduced a decade's worth of work to two or three days. After reviewing the pertinent reactions, the two researchers determined that most of the chlorine atoms combine with ozone and form the free radical chlorine oxide, which in turn becomes part of a chain reaction. A single chlorine atom can remove as many as 100,000 molecules of ozone.

In 1974, Rowland and Molina made a disturbing prediction: If industry continued to release a million tons of CFCs into the atmosphere each year, atmospheric ozone would eventually drop by 7 to 13 percent.

Mitigation Measures

- Montreal Protocol – adopted in 1987 and enforced in 1989
- All developed nations stopped production & Consumption of ODSs by 1996.
- For developing nations it is by 2010
- Montreal protocol compels the countries producing ODS > 300 mg/capita/year to fund the fight against the O_3 depletion

- Consumption of ODS in India: 10 mg/capita/year
- Hence India is receiving special funds to switch over to CFC-free technologies

Alternatives for dangerous CFCs

- HFCs (Hydrofluorocarbons) – for refrigeration applications
- HCFCs (Hydrochlorofluorocarbons) – for cleaning and foam manufacture applications
- These will not reach the stratosphere as they contain H atoms (which react with hydroxyl radicals of the lower atmosphere lowering their life span)
  - These compounds are global warming gases
  - These compounds will generate more O3 in the lower atmosphere causing smog

Repairing the O_3 holes

- Releasing large quantities of ethane and propane in the stratosphere
- These compounds can bind to the free Cl preventing them from attacking O_3 layer.

- September 16th: international day for the preservation of O_3 layer – UN general assembly

- CFC release has reduced considerably
- Troposphere Cl peaked in late nineties
- Stratospheric Cl must have peaked by 2004
- Ozone layer depletion will continue for a few more decades.
- By 2050, no ozone layer depletion will be there if all recommendations (Montreal protocol) are implemented in its tue spirit.